KIRAN DAMODARAN

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SUMMARY

I build AI systems that actually work in the real world, not just in notebooks. At Zenon Robotics, I lead the full AI stack for Robin, a production robot deployed in the UAE. My work spans computer vision, multilingual LLM pipelines, edge inference on NVIDIA Jetson, and everything in between. Before that, I spent a year at Colorado School of Mines turning raw methane sensor data into a 99%-accurate anomaly detection system that runs in production today.

TECHNICAL SKILLS

Languages: Python, R, SQL, Dart, JavaScript, C#
AI / ML: TensorFlow, PyTorch, Scikit-learn, LLMs, CNNs, YOLO, Isolation Forest, Anomaly Detection
CV & NLP: OpenCV, Tesseract OCR, Multimodal Pipelines, Fine-Tuning, Prompt Engineering, STT/TTS, NVIDIA Jetson
MLOps & DevOps: Model Drift Monitoring, Regression Testing, Docker, CI/CD, GitHub Actions, AWS, Linux
APIs & Data: FastAPI, REST APIs, WebSockets, SQLite, ETL/ELT, Snowflake, IBM DataStage, QGIS, pandas

EXPERIENCE

AI Engineer | Zenon Robotics & Smart Machines — Jan 2026 – Present | Dubai, UAE

- Built a 4-mode Scan System (AUTO/OCR/ID/ONESHOT) on Robin pet robot using Tesseract OCR, Vision LLM API, and OpenCV, hitting 97% document and ID recognition accuracy in live deployment.
- Designed a multilingual AI conversation pipeline (English, Arabic, Farsi) using Vosk STT, Zenon LLM API, and Piper TTS with intent routing, achieving full response cycle under 2 seconds.
- Shipped a YOLO-based object detection and follower control system that tracks targets in real time and drives robot wheel velocity directly from camera feed, reducing tracking lag by 40%.
- Built a 60-motion library (gestures, dance, yoga, navigation patterns) and a 50-expression facial animation engine coordinated with the robot speech output for natural interaction.
- Packaged the full ROS2 AI stack as 8 independently deployable lifecycle nodes on NVIDIA Jetson GPU, improving system responsiveness by 55% and enabling per-service hot restarts.
- Reduced robot commissioning time by 70% by building a zero-touch QR and Wi-Fi provisioning pipeline and a Flutter mobile app with 9 live control panels used daily by the ops team.

Research Scientist | Colorado School of Mines, Payne Institute — Jun 2024 – Jun 2025 | Colorado, USA

- Designed a spike-detection algorithm for time-series methane sensor data across 50 METEC sensors, achieving 99% anomaly detection accuracy validated against ground-truth emissions events.
- Built Isolation Forest pipelines paired with LLM-based root-cause analysis, cutting the time from alert to actionable insight by 30% on high-frequency sensor streams.
- Reduced geochemical ML forecasting compute time by 85% through R workflow refactoring and feature engineering and built a TDD test suite that caught 10 critical bugs before production.
- Engineered a Python dead-letter queue pipeline for fault-tolerant sensor ingestion, processing 60% more data per hour without data loss across multi-network sensor arrays.

Associate Data Analyst, AI and Analytics | Cognizant — Jun 2021 – Jul 2022 | Chennai, India

- Built optimized ETL pipelines in IBM InfoSphere DataStage feeding RBC's ML-ready data warehouse, increasing clean data throughput by 25% across millions of daily transactions.
- Automated JCL transaction validation to enforce 90% data integrity on inputs to downstream analytics models, and designed Informatica CI/CD pipelines cutting deployment errors by 20%.

EDUCATION

M.S. Computer Engineering — GPA 3.71 | Colorado State University — Aug 2022 – May 2024 | USA

PROJECTS

Embedded Network and Security Research | Colorado State University — Sep 2022 – May 2024

- Improved network uptime by 45% by deploying QGIS telemetry diagnostics across 50 embedded nodes on AT&T, Verizon, and CSU-NET, and built an ABAC security framework cutting vulnerabilities by 40%.

Vehicle Type and Color Classification via CNN | Colorado State University — May 2023

- Fine-tuned pre-trained TensorFlow CNN models on 10,000 labeled vehicle images, reaching 94% accuracy on multi-class type and color classification for traffic analysis.